1. Let
$$A = \begin{pmatrix} 3 & -1 & 2 \\ 0 & 4 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

   a. What is $\det(A^4/3)$?

   b. What is A^{-1}

2. A ball is dropped and its speed is measured at the n times t_i, where $i = 1, \ldots, n$. This yields the data (t_i, v_i). The ball is expected to fall with the constant acceleration a, meaning that its speed is expected to satisfy
$$v(t) = v_0 - at.$$
Least-squares regression can be used to fit this model to the data by solving
$$M^T M \begin{pmatrix} v_0 \\ -a \end{pmatrix} = M^T \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}.$$
In this case,
$$M^T M = 200 \begin{pmatrix} 1 & 5 \\ 5 & 30 \end{pmatrix}.$$

   a. How would M be constructed from the data?

   b. What is the average measured time $\langle t \rangle = \sum_{i=1}^{n} t_i / n$?

3. Find c such that
$$B = \begin{pmatrix} 7 & 2 & 1 \\ 1 & c & 2 \\ 2 & 0 & 1 \end{pmatrix}$$
cannot be inverted